

DIMITRIS SPATHOULAS

M.Sc. in Artificial Intelligence · KU Leuven
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EDUCATION

KU Leuven

Leuven, Belgium

M.Sc. in Artificial Intelligence. *Magna Cum Laude*, 79.83/100.

Sep. 2025 – Sep. 2026

Engineering and Computer Science Track.

Master Thesis: Projected Subgradient Optimization with Polyak Steps and Langevin Perturbations for Constrained Single-Cell Representation Learning under Non-Smooth Potentials [1].

Description: Formulated a non-negative autoencoder architecture using projected subgradient methods for representation learning, and investigated discretized Langevin dynamics to sample posterior models under non-smooth prior objectives.

Research Focus: Optimization, Stochastic Modeling, SDEs, MCMC, Probability and Statistics, Representation Learning.

Related Coursework: Artificial Neural Networks and Deep Learning, Uncertainty in AI, Reinforcement Learning, Support Vector Machines.

Democritus University of Thrace

Xanthi, Greece

M.Sc. in Production Engineering and Management. *Summa Cum Laude*, 8.71/10.

Sep. 2019 – Jan. 2025

Integrated Master's, 300 ECTS, 5-year program.

Master Thesis: 3D Multi-modal Multi-object Tracking via Machine Learning and Analytic Collision Risk Calculation for Autonomous Vehicles Navigation [2, 3, 4, 5].

Description: A real-time feature extraction and multi-modal fusion pipeline for spatiotemporal correlation of target entities, processing over 300GB of real-world data across 7 sensors (1 LiDAR, 6 cameras) and an analytic collision risk module computing overlaps between geometric safety zones and Gaussian-distributed obstacle tracks.

Research Focus: 3D Computer Vision, Multimodal Learning, Large-Scale Data Processing, Dynamics, Robotics.

Related Coursework: Systems and Control, Numerical Analysis, Decision Theory, Stochastic Processes.

CONFERENCES AND GUEST LECTURES

39th Conference of Hellenic Mathematical Society

Olympia, Greece, 2024

Main Track Presentation Title: Neural Networks: The Foundational Cornerstone of Modern Machine Learning

Demonstrated the mathematical analysis of neural computation, focusing on perceptron mechanics and cross-entropy binary classification.

Structured Programming

Democritus University of Thrace, Greece, 2024

Lecture topic: Linear Systems, Computer Vision and Robotics in C

Independently developed and delivered a full-length lecture in the course *Structured Programming*, after approval from Prof. Dr. Stefanos Katsavounis, presenting original material.

SKILLS

Programming & Technical Computing: Python, C/C++

Deep Learning & Scientific Computing: PyTorch, NumPy, scikit-learn, TensorFlow, SciPy, SymPy, FilterPy, scanpy

Development, Tools & Methods: Git, Anaconda

Operating Systems & Middleware: Linux (Ubuntu, WSL2), SSH

Languages: Greek, English

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Summary of Theses

PART I: CURRENT MASTER’S THESIS

Title: Projected Subgradient Optimization with Polyak Steps and Langevin Perturbations for Constrained Single-Cell Representation Learning under Non-Smooth Potentials

Objective and Formulation

The project formulates the learning of low-dimensional representations from high-dimensional single-cell RNA sequencing data as a constrained, non-smooth optimization problem. The architecture is a linear autoencoder with tied, non-negative weights. The objective function $\mathcal{J}(W)$ integrates the reconstruction loss $\mathcal{L}(A, \hat{A})$ (which can be computed using either the squared Frobenius norm or the generalized Kullback-Leibler divergence), an ℓ_1 sparsity penalty, a squared Frobenius norm penalty ($\|W\|_F^2$), and optionally a Laplacian graph-based regularization:

$$\min \mathcal{J}(W) = \mathcal{L}(A, \hat{A}) + \lambda_{\text{graph}} \mathcal{L}_{\text{graph}} + \lambda_1 \|W\|_1 + \lambda_2 \|W\|_F^2 \quad \text{s.t.} \quad A, \hat{A} \geq 0, W, Z \geq 0$$

Which is then decomposed into a smooth component, $h(W)$, and a non-smooth component, $c(W)$. The generalized subgradient at epoch k is then:

$$G^{(k)} = \underbrace{\nabla_W \mathcal{L}(A, \hat{A}) + \lambda_{\text{graph}} \nabla_W \mathcal{L}_{\text{graph}} + 2\lambda_2 W^{(k)}}_{\nabla h(W^{(k)})} + \underbrace{\lambda_1 \text{sgn}(W^{(k)})}_{\in \partial c(W^{(k)})}$$

where sgn is the signum function applied element-wise to each weight $W_{ij}^{(k)}$.

Deterministic Optimization Dynamics

Due to the non-differentiability of the ℓ_1 penalty at the origin and the hard non-negativity constraints, the study evaluates projected subgradient descent strategies. Among the four rules tested—constant, diminishing, decaying, and scaled Polyak—the scaled Polyak step-size achieves superior convergence and clustering performance:

$$\alpha_k = \alpha \frac{\mathcal{J}(W^{(k)}) - \mathcal{J}^*}{c_{po} \|G^{(k)}\|_F^2}$$

Where α, c_{po} scaling factors.

Stochastic Exploration via Langevin Dynamics

Langevin dynamics are introduced to explore the posterior parameter space via the Unadjusted Langevin Algorithm, combined with Euclidean projections onto the feasible set:

$$W^{(k+1)} = \Pi_{\geq 0}(0, W^{(k)} - \alpha_k G^{(k)} + \sqrt{2\alpha_k} \xi^{(k)})$$

where $\xi^{(k)}$ is a matrix with each element drawn i.i.d. from $\mathcal{N}(0, 1)$.

To approximate the Bayesian Model Average without excessive memory overhead, an online ergodic average of the network weights is maintained. The averaged weights W_{Lang} are updated incrementally at each sampled step s according to:

$$W_{\text{Lang}}^{(s)} = \frac{s-1}{s} W_{\text{Lang}}^{(s-1)} + \frac{1}{s} W^{(k_s)}$$

The Langevin perturbations conflict with non-negative projections, preventing exact sparsity and inflating parameter norms due to the positive drift created. While this limits stochastic refinement in linear models, it still marginally improves clustering on higher-dimensional datasets. Ultimately, the scaled Polyak step-size rule yields the main performance advantages when compared to standard PCA, NMF, and Adam-based optimization.

Title: Real-time 3D Multi-modal Multi-object Tracking via Machine Learning and Analytic Collision Risk Calculation for Autonomous Vehicles Navigation

State Estimation and Feature Fusion

The system performs 3D multi-object tracking utilizing data from one LiDAR and six cameras. State estimation was formulated via a 3D Kalman Filter using a constant linear and angular velocity dynamic model. The process noise covariance and observation noise covariance matrices were computed directly from the variance of sequential measurements. The state vector consists of:

$$s_t = (x, y, z, a, l, w, h, dx, dy, dz, da)^T$$

where x, y, z denote the target's position in the reference frame, a its orientation, l, w, h the dimensions of the bounding box, and dx, dy, dz, da the change in these parameters between two consecutive detections.

The geometric features (from LiDAR, $F_{3D} \in \mathbb{R}^{N \times 512 \times 3 \times 3}$) and appearance features (from the corresponding camera out of the 6, $F_{2D} \in \mathbb{R}^{N \times (1024+6)}$) are fused into a joint representation $F_{det}^{fused} \in \mathbb{R}^{N \times 512 \times 3 \times 3}$ using a learned multi-layer perceptron $G_1(\cdot)$. This unit serves a twofold purpose: it transforms the dimensions of the appearance features to match the geometric features, and it aligns their vectors directionally so that their pairwise addition becomes meaningful:

$$F_{det}^{fused} = G_1(F_{2D}) + F_{3D}$$

These fused features for incoming detections and existing tracks (F_{trk}^{fused}) are then processed by a secondary neural network $G_2(\cdot)$ to compute a deep affinity matrix D_{feat} , expressing the pairwise similarities for data association:

$$D_{feat} = G_2(F_{det}^{fused}, F_{trk}^{fused})$$

To address the sparsity of the true association matrix during training, the two networks are optimized using a focal loss formulation, which down-weights the overwhelming number of negative (non-matching) associations:

$$FL(p_t) = -\alpha \cdot (1 - p_t)^\gamma \cdot \log(p_t)$$

where p_t represents the predicted probability of the correct class, α is the ratio between negative and positive associations, and γ is a focusing parameter. Empirical results show that this deep affinity matrix is more representative when compared to simpler standard techniques such as cosine similarity.

Analytic Collision Risk Calculation

The Collision State Probability was calculated by modeling the spatial overlap between the autonomous vehicle and tracked obstacles. The Minkowski sum of the object bounding geometries was computed to produce a collision octagon. The coordinate system is transformed based on the obstacle's position and orientation, so that the principal axes of the obstacle's Gaussian covariance align with the orthogonal axes of the reference frame. The collision risk was evaluated analytically by integrating over the independent Gaussian state distributions of the objects within this polygonal domain. By decomposing the boundary into regions of positive and negative contribution, the total probability at time T_p was computed using the Gaussian error function (erf):

$$P(C, T_p) = \sum_{i=0}^{i \leq m} P(C_i, T_p) - \sum_{i=m}^{i \leq n} P(C_i, T_p)$$

where $P(C_i, T_p)$ represents the probability contribution of an individual line segment C_i of the collision octagon. By decomposing the octagon into these line segments, the contribution is calculated by numerically integrating the Gaussian error function over the boundaries of the segment. Line segments are assigned a positive or negative contribution depending on whether they lie below or above the horizontal axis, while segments nearly parallel to the vertical axis are excluded. If the octagon lies below the x -axis, the positive and negative contributions are reversed to correctly capture the enclosed probability volume:

$$P(C_i, T_p) = \frac{1}{\sqrt{8\pi}\sigma_x} \int_{x_{l,i}}^{x_{u,i}} \text{erf}\left(\frac{m_i x + b_i}{\sqrt{2}\sigma_y}\right) e^{-\frac{x^2}{2\sigma_x^2}} dx$$



TRANSCRIPT

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AE4DF0905CAB48869E451199A7F68836

Student number: 1083798

Dimitrios Spathoulas, born in Thessaloniki (Greece) on 09.04.2001

Programme: Master of Artificial Intelligence (Leuven) - Master of Science

Language of instruction: EN

Course Units			Language Credits (ECTS)	Grade	C/T/V (*)
<i>Academic year 2025-2026</i>					
Privacy and Big Data	H00Y2A	EN	4	14	C
Fundamentals of Artificial Intelligence	H02A0A	EN	5	15	C
Constraint Solving	H02A3A	EN	4	16	C
Computer Vision	H02A5A	EN	4	19	C
Machine Learning and Inductive Inference	H02C1A	EN	4	17	C
Artificial Neural Networks and Deep Learning	H02C4A	EN	4	16	C
Uncertainty in Artificial Intelligence	H02D2A	EN	4	12	C
Support Vector Machines: Methods and Applications	H02D3A	EN	4	17	C
Philosophy of Mind and Artificial Intelligence	H02D5A	EN	4	19	C
Master's Thesis ECS	H02D6A	EN	15	17	C
Cybernetics and its Applications in Physiology and Biological Sciences	H02H5A	EN	4	15	C
Reinforcement Learning	H0023A	EN	4	12	C

Overall result: Passed Magna cum laude with 79,83 %

Supplementary titles

H02D6A Projected Subgradient Optimization with Polyak Steps and Langevin Perturbations for Constrained Single-Cell Representation Learning under Non-Smooth Potentials

Date of issue of results: 11.09.2026

Rector

Prof. dr. Severine Vermeire



Student number: 1083798

Dimitrios Spathoulas, born in Thessaloniki (Greece) on 09.04.2001

Programme: Master of Artificial Intelligence (Leuven) - Master of Science

EXPLANATORY NOTES

Course units at KU Leuven are evaluated on a grading scale of 0 to 20. As an exception it is also possible for certain course units, or parts of course units, to have a pass/fail evaluation. The Flemish credit system is fully in accordance with the European Credit Transfer and Accumulation System (ECTS).

Overview of the codes and abbreviations used on the transcript:

C	The student has passed the course unit, and has therefore obtained a credit.
T	The student has not passed the course unit, but the fail mark was tolerated.
V	The student has been exempted for the course unit.
NA	The student has not taken the exam for the course unit.
GR	There is no grade for the course at this point, but a grade is expected.
G/P	The student has obtained a "pass" grade for a course unit for which a pass/fail evaluation was employed.
NG/F	The student has obtained a "fail" grade for a course unit for which a pass/fail evaluation was employed.
NVT	This part of the course unit is not relevant for the student.
NL	This course unit was offered in Dutch.
EN	This course unit was offered in English.
FR	This course unit was offered in French.
ES	This course unit was offered in Spanish.
DE	This course unit was offered in German.

Descriptions of the course units can be consulted in KU Leuven's online archive of its programmes, on <https://onderwijsaanbod.kuleuven.be/archief>.

A description of the grading system can be found on <http://www.kuleuven.be/english/education/student/examinations/grading-system>. Information on the positioning of the exam results can be found on <https://www.kuleuven.be/english/education/ects/positioning>.



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STUDENT NUMBER 1083798

LEUVEN 2026-09-11

Degree certificate

The rector of KU Leuven

hereby testifies that

Dimitrios Spathoulas

born in Thessaloniki on 09.04.2001

did on 11.09.2026

receive the following degree

Master of Artificial Intelligence (Master of Science: MSc)

Magna cum laude.

Specialisation: Engineering and Computer Science.

This certificate is issued at the request of the diploma bearer and functions as a legal substitute for the diploma.

On behalf of the rector

A handwritten signature in black ink, belonging to Jan Raeymaekers.

Jan Raeymaekers
Director Teaching and Learning Processes

This certificate with signature has been generated electronically on the basis of a verified electronic file.



DEMOCRITUS UNIVERSITY OF THRACE

SCHOOL OF ENGINEERING

DEPARTMENT OF PRODUCTION AND MANAGEMENT ENGINEERING

null

Tel.: +302541079360

www.pme.duth.gr

DIPLOMA SUPPLEMENT**(Valid with diploma No 1435)**

This Diploma Supplement follows the model developed by the European Commission, Council of Europe and UNESCO/CEPES. The purpose of the supplement is to provide sufficient independent data to improve the international 'transparency' and fair academic and professional recognition of qualifications (diplomas, degrees, certificates etc.). It is designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It should be free from any value judgements, equivalence statements or suggestions about recognition. Information in all eight sections should be provided. Where information is not provided, an explanation should give the reason why.

1. INFORMATION IDENTIFYING THE HOLDER OF THE QUALIFICATION**1.1 Family Name(s): SPATHOULAS****1.2 Given name(s): DIMITRIOS****1.3 Date of birth (day/month/year): 09/04/2001****Place, Country of Birth: THESSALONIKI, GREECE****1.4 Student identification number or code (if available): 92206****2. INFORMATION IDENTIFYING THE QUALIFICATION****2.1 Name of qualification and (if applicable) title conferred (in original language):**

ΔΙΠΛΩΜΑ ΜΗΧΑΝΙΚΩΝ ΠΑΡΑΓΩΓΗΣ ΚΑΙ ΔΙΟΙΚΗΣΗΣ

2.2**2.3 Name and status of awarding institution (in original language):**ΔΗΜΟΚΡΙΤΕΙΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΘΡΑΚΗΣ, transliterated into: Dimokriteio Panepistimio Thrakis
- in English: Democritus University of Thrace, D.U.Th., Public University**2.4 Name and status of institution (if different from 2.3) administering studies (in original language):**

As in 2.3

2.5 Language(s) of instruction/examination:

GREEK

3. INFORMATION ON THE LEVEL OF THE QUALIFICATION**3.1 Level of qualification:**

Integrated Master

3.2 Official length of programme:

5 years / 10 semesters, 300 ECTS

3.3 Access requirement(s):

Certificate from High School (Secondary Education) and examinations. Persons subject into

specific categories and Degree holders of other Universities, (3%) after placing examinations.

4. INFORMATION ON THE CONTENTS AND RESULTS GAINED

4.1 Mode of study:

Full-time 5 years studies.

4.2 Programme requirements:

According to the National Law for Higher education and the Department's Internal Regulation, students graduate after having:

- successfully attended all courses (56) of the programme
- successfully completed, defended and examined in the final Diploma Dissertation
- accumulated 300 ECTS credits.

Upon completion of the studies, the graduates acquire the necessary scientific knowledge, skills and competences on the field of Production and Management Engineering and have the opportunity to pursue their professional development.

The graduates will be able to:

Practice the profession of Production and Management Engineering.

4.3 Programme details (e.g. modules or units studied) and the individual grades/marks/credits obtained:

The courses in which the above mentioned student has been examined and got passing grades, as well as courses for which he/she has received recognition, are the following (C=Compulsory, O=Optional, OD,O1,O2=Optional Direction, CS=Compulsory Selective, CD=Compulsory Direction, S=Specialization):

Course Code	Course Title	Course Type	Grade	Examination period	Examination Type	ECTS
A1Y	Linear Algebra and Discrete Mathematics	C	8	FEB 2019-2020	Examination	5
A2Y	Introduction to Computing	C	9.5	FEB 2019-2020	Examination	5
A3Y	Calculus I	C	6	FEB 2019-2020	Examination	5
A7Y	Management for Engineering	C	6	FEB 2019-2020	Examination	5
A9Y	General Physics	C	5	SEPT 2019-2020	Examination	5
B11Y+	Mechanical Design I	C	7.5	FEB 2019-2020	Examination	5
B5Y	Structured Programming with C	C	9.5	JUNE 2019-2020	Examination	5
Γ4Y	Applied Thermodynamics	C	6	JUNE 2019-2020	Examination	6
B10Y+	Operations Research	C	7	JUNE 2019-2020	Examination	5
B12Y	Research Methodology	C	10	SEPT 2019-2020	Examination	5
B1Y	Probability and Statistics	C	6	JUNE 2019-2020	Examination	4
B2Y	Calculus II	C	10	SEPT 2023-2024	Examination	5
Γ5Y	Mechanics I (Statics)	C	9.5	SEPT 2023-2024	Examination	6
Γ2Y	Object - Oriented Programming with C++	O	9.5	FEB 2020-2021	Examination	3
Γ3Y	Technology of Materials I	C	10	FEB 2022-2023	Examination	5
Θ02	Product Design	C	5	SEPT 2022-2023	Examination	5
Z06	MARKETING	C	6.5	FEB 2020-2021	Examination	5
ΣΤ9Y	Heat Transfer	C	10	FEB 2020-2021	Examination	6

Course Code	Course Title	Course Type	Grade	Examination period	Examination Type	ECTS
ΣΤ3Ε	Numerical Analysis	O	10	JUNE 2020-2021	Examination	3
Δ4Υ	Information Systems	C	7	JUNE 2020-2021	Examination	5
Δ7Υ	Production Systems	C	5	JUNE 2020-2021	Examination	6
Δ1Υ	Differential Equations	C	8.5	SEPT 2021-2022	Examination	5
Δ2Υ+	General Electrotecnics	C	8.5	JUNE 2020-2021	Examination	6
Δ9Υ	Mechanics II (Strength of Materials)	C	6.5	JUNE 2020-2021	Examination	5
E3Υ	Fluid Mechanics	C	9	FEB 2021-2022	Examination	6
E4Υ	Production Technology	C	6	FEB 2021-2022	Examination	5
E8Υ	Mechanics III (Dynamics)	C	10	FEB 2021-2022	Examination	6
E1Ε	Electrical Machines	C	10	FEB 2021-2022	Examination	5
ΕΠ8	ALGORITHMIC OPTIMIZATION METHODS	O	10	FEB 2021-2022	Examination	3
ΣΤ10Υ	Industrial and Technological Marketing	C	5	FEB 2021-2022	Examination	5
ΕΠ9	Geometric Transformations	O	10	JUNE 2021-2022	Examination	3
ΣΤ2Υ	Electronics	C	10	JUNE 2021-2022	Examination	6
Δ4Ε+	Human Resource Management	C	8.5	JUNE 2021-2022	Examination	5
E1AY	Mechanical Design II	C	6	JUNE 2021-2022	Examination	6
ΣΤ7Υ	Systems of Industrial Processes	C	5	SEPT 2023-2024	Examination	5
H07	Industrial Electrical Installations	C	7	JUNE 2021-2022	Examination	5
Z13	Mechatronics	C	9	FEB 2022-2023	Examination	5
H04	Decision Support Systems	C	9	FEB 2022-2023	Examination	5
Z01	Control Systems	C	10	SEPT 2022-2023	Examination	6
Z07	Information Systems Management	C	10	FEB 2022-2023	Examination	6
Θ04	Project Management	C	10	FEB 2022-2023	Examination	5
Θ08	Intelligent Manufacturing Systems	O	10	FEB 2022-2023	Examination	3
ΕΠ3	Computer Vision	O	10	JUNE 2022-2023	Examination	3
H15	Propulsion Systems	C	10	JUNE 2022-2023	Examination	6
Z11	Engineering Economics	C	7	SEPT 2022-2023	Examination	5
H03A	Environmental Engineering	C	7.5	JUNE 2022-2023	Examination	5
H14Ε	Computational Intelligence	C	8	JUNE 2022-2023	Examination	5
H01	Robotics	C	10	JUNE 2022-2023	Examination	6

Course Code	Course Title	Course Type	Grade	Examination period	Examination Type	ECTS
H10	Fluid Mechanics Engines	C	10	FEB 2023-2024	Examination	5
H12	Inventory Management	C	9.5	FEB 2023-2024	Examination	5
H13	Supply Chain Management	C	10	FEB 2023-2024	Examination	5
Z14Y	Mental ergonomics and interactive design	C	10	FEB 2023-2024	Examination	4
Θ10	Industrial Electronics and Automation	O	10	FEB 2023-2024	Examination	3
ΣT4E	Stochastic Processes	O	10	FEB 2023-2024	Examination	3
Θ07	Technological Innovation and Entrepreneurship	C	8.5	FEB 2023-2024	Examination	5

Courses ECTS Total (1)
270

The column of ECTS presents the credits that correspond to each course.

Exemption is defined as recognition applied when a student carries the ECTS credits from previously completed and recognized studies, e.g. Erasmus.

*Does not count towards diploma.

Diploma Thesis Grade: 10

Diploma Thesis ECTS (2): 30

ECTS Total (1)+(2): **300**

4.4 Grading scheme and, if available, grade distribution guidance:

According to the Institution's Internal Regulations, the grading system falls into the 0-10 scale as follows:

8,50-10,0: " Excellent "

6,50-8,49: " Very Good "

5,00-6,49: " Good "

0,00-4,99: " Fail "

At least a grade of 5.0 is required for the successful completion of a course.

4.5 Overall classification of the qualification:

"EXCELLENT" 8,71 EIGHT AND SEVENTY-ONE HUNDREDTHS

5. INFORMATION ON THE FUNCTION OF THE QUALIFICATION

5.1 Access to further study:

Access to Post Graduate studies in Greece or abroad.

5.2 Professional status (if applicable):

According to the relevant laws.

6. ADDITIONAL INFORMATION

6.1 Additional information:

6.2 Further information sources:

Democritus University of Thrace:

<http://www.duth.gr>

Democritus University of Thrace, Department of Production and Management Engineering:

<http://www.pme.duth.gr>

Hellenic Ministry of Education, Lifelong Learning and Religious Affairs:

<http://www.minedu.gov.gr>

7. CERTIFICATION OF THE SUPPLEMENT

7.1 Date: 26/05/2025

7.2 Name and Signature:

George Gaidajis
Professor

Sofia Gkazi

Georgios Gkaintatzis

03.06.2025 10:23 UTC

Head of the Department



Digitally signed by Sofia
Gkazi

Date: 2025.05.27 10:53:24
EEST

Deputy Secretary of the Department

7.3 Capacity:

7.4 Official stamp or seal:

8. INFORMATION ON THE NATIONAL HIGHER EDUCATION SYSTEM

(i) Structure:

According to the Framework Law (2007), higher education consists of two parallel sectors: the University sector (Universities, Polytechnics, Fine Art Schools, the Open University) and the Technological sector (Technological Education Institutions (TEI) and the School of Pedagogic and Technological Education). The same law regulates issues concerning governance of higher education along the general lines of increased participation, greater transparency, accountability and increased autonomy.

There are also state non-university tertiary Institutions offering vocationally oriented courses of shorter duration (two to three years) which operate under the authority of other ministries.

(ii) Access

Entrance to the various Schools of the universities (Panepistimio) and technological education institutions (Technoligo Ekpaideftiko Idryma - TEI) depends on the general score obtained by Lyceum graduates on the certificate, as described above (Section 5.iv), on the number of available places (numerous clausus) and on the candidates' ranked preferences among schools and sections.

(iii) Qualifications

Students who successfully complete their studies at Universities and TEI are awarded a Ptychio (first cycle degree). First cycle programmes last four years for most fields to five years for engineering and certain other applied science fields and six years for medicine. The Ptychio leads to employment or further study at the post-graduate level that includes the one year second cycle leading to the second degree, Metaptychiako Dimpoma Eidikefsis - equivalent to the Master's degree - and the third cycle leading to the doctorate degree, Didactoriko Diploma.

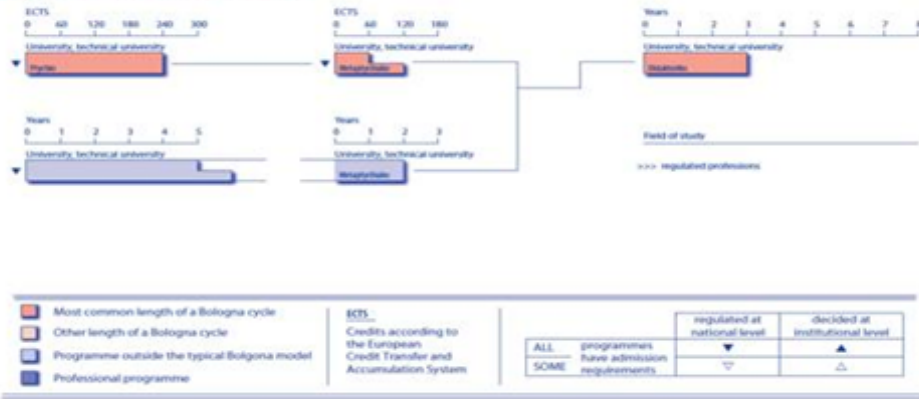
Recent legislation on quality assurance in higher education, the credit transfer system and the diploma supplement defines the framework and criteria for evaluation of university departments and for certification of student degrees. These measures aim at promoting student mobility and contributing to the creation of a European Higher Education Area.

A detailed description of the Greek Education System is offered in the EURYDICE <http://www.eurydice.org> database of the European Education Systems.

http://eacea.ec.europa.eu/education/eurydice/documents/thematic_reports/122EN.pdf (pages 82,83)

<http://www.eurydice.org>

Higher education structure – 2010



DIMITRIS SPATHOULAS

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REFERENCES

Prof. Dr. Yves Moreau

Full Professor, Department of Electrical Engineering (ESAT)
STADIUS, KU Leuven

Email: yves.moreau@kuleuven.be

Relation: Supervisor for M.Sc. Artificial Intelligence Thesis

Prof. Dr. Antonios Gasteratos

Dean, School of Engineering & Director, Robotics and Automation Lab
Democritus University of Thrace

Email: agaster@pme.duth.gr · **Phone:** +30 25410 79359

Relation: Supervisor for 5-year Integrated Master's Thesis

Dr. Susan Ghaderi

FWO Special Research Associate (Postdoctoral)
STADIUS, KU Leuven

Email: susan.ghaderi@kuleuven.be

Relation: Daily Advisor for M.Sc. Artificial Intelligence Thesis